

Week 1: Project Proposal

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Project Pitch

Game Title: Epic Brawlers

Description:

Epic Brawlers is a click-to-fight boss game built with Java Swing. The player faces one enemy boss at a time. Clicking **attack** inflicts damage; defeating a boss awards gold, which the player can spend on **weapon upgrades** to increase damage per click. Selecting to move onto the **Next Boss** spawns a tougher opponent with higher health and larger rewards.

All gameplay happens through GUI components — buttons, labels, and a progress bar. There's no player movement; instead, progression focuses on stages, upgrades, and incremental strength.

Gameplay each round:

When the player clicks Attack, the player deals damage to the boss, and immediately after, the boss attacks back. This ensures that both sides lose HP each round. When the boss's HP reaches 0, the player earns gold and has the option to upgrade their weapon (upgrades unavailable if funds are insufficient) or advance to the next stage, where the next boss has higher HP, greater attack damage, and better rewards.

User actions:

- Clicks **Attack** —> Boss HP decreases. If $HP \leq 0$, the boss dies, the player earns gold, and a new boss appears.
- Clicks **Upgrade Weapon** —> if enough gold, the player's weapon level increases, raising its attack damage
- Clicks **Next Boss** —> Generates the next-tier boss.
- **Stage Progression** —> defeating a boss allows to advance to the next stage
- **Death/Reset** —> if $HP \leq 0$, the player restarts at Stage 1, fully healed.

GUI Preview

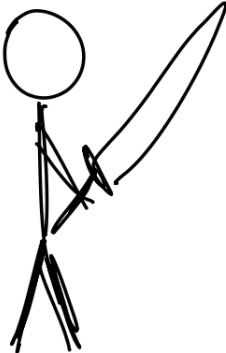
Stage 1

Player HP

Boss HP

Gold

Level





Attack

Upgrade Wep

Next Boss

CRC Cards

Player: tracks gold, level, damage, hp; handles upgrades and taking damage.

Collaborator(s): Boss

Boss: stores hp, maxHp, reward, attack, stage; handles taking damage

Collaborator(s): Player

GameUI: manages GUI display and logic; responds to button clicks; updates HP bars and stage progression

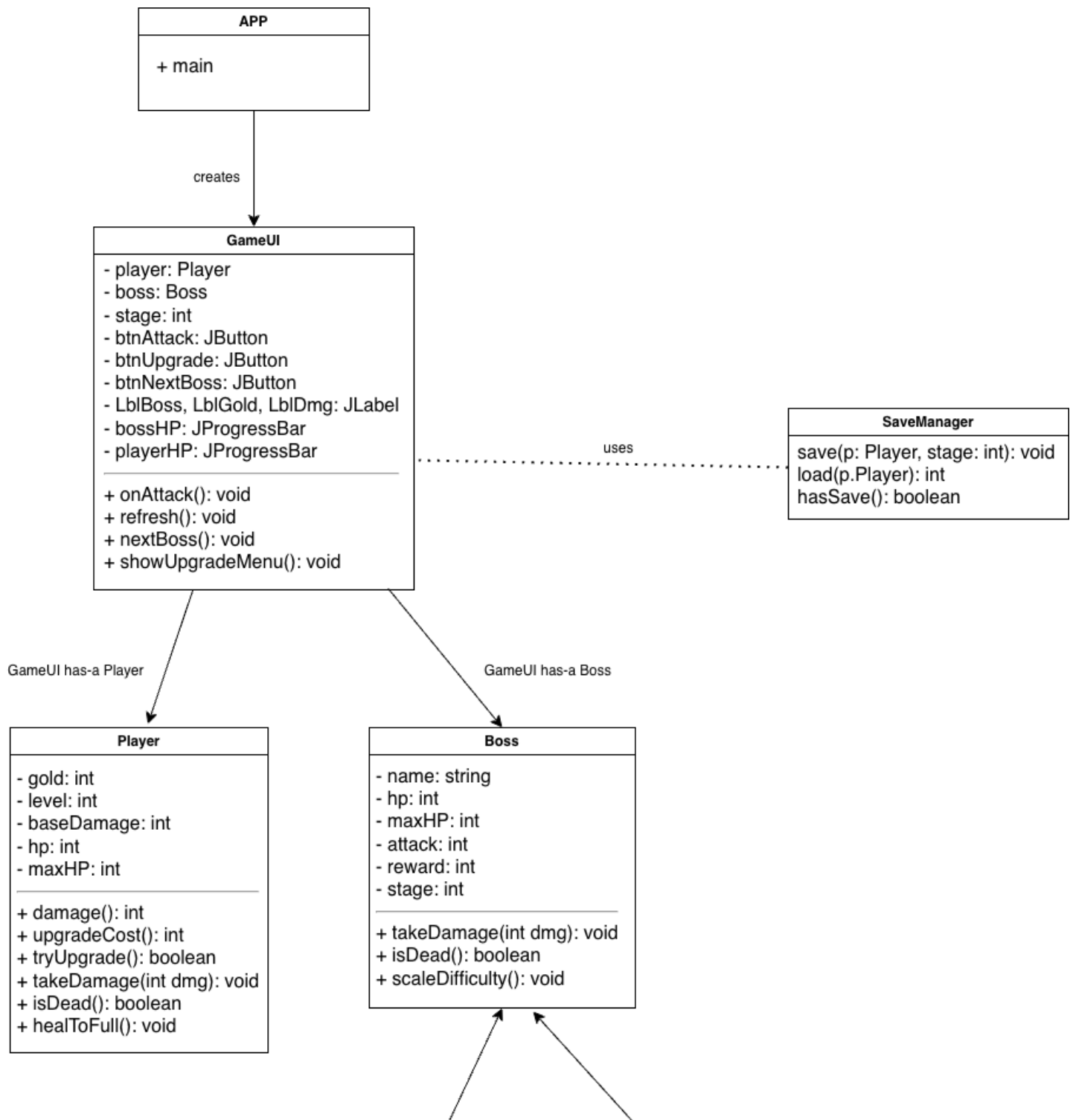
Collaborator(s): Player, Boss

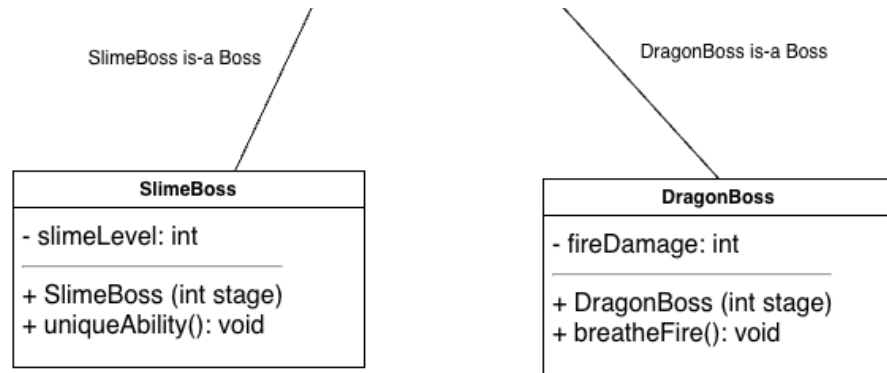
***BaseWeapon:** base class for all weapons; defines common fields (name, damage, level) and methods for attacking and upgrading

Collaborator(s): *Player, Boss*

*= possibly added in future

UML Diagram:





Object-Oriented Design

- Aggregation: The GameUI has-a Player and a Boss
- Encapsulation: Each class manages its own state and behavior (damage, HP, gold).
- Polymorphism: future versions include subclasses such as a SlimeBoss and DragonBoss to demonstrate inheritance
- Event-Driven Programming: each click triggers ActionListener events, updating the GUI and progressing the game round-by-round

Video Link: https://youtu.be/fAck9a8_tyl

Learning Outcomes

LO	Demonstrated in Epic Brawlers
LO1: Object Oriented Design Principles	Clear separation of classes; encapsulated Player/Boss classes; GUI/controller in GameUI class
LO2: · LO2: Construct programs utilizing single and multidimensional arrays (optional)	Optional
LO3: Objects & Aggregation	GameUI aggregates Player and Boss and coordinates their interactions
LO4: Inheritance & polymorphism	Planned boss hierarchy (e.g., BaseBoss —> SlimeBoss, DragonBoss) without changing controller code
LO6: GUI & Event-Driven Programming	Swing (JFrame, JButton, JLabel, JProgressBar) with ActionListener callbacks for attacks/upgrades
LO7: Exception handling	Handling for insufficient funds on upgrade; safe HP bounds; death/reset flow
LO8: Text File I/O	Save/load player progress (gold, level) to a simple text file

Planned Working Time

Week	Hours (outside class)
Week 1	Thurs 2-5 PM + Fri 6-9 PM
Week 2	Thurs 2-5 PM + Fri 6-9 PM
Week 3	Thurs 2-5 PM + Fri 6-9 PM / Sat 11 AM - 2 PM
Week 4	Thurs 2-5 PM + Fri 6-9 PM / Sat 11 AM - 2 PM
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TO-DO Timeline Plan

Week	Goals
1	Write proposal; design CRC cards and UML; set up project files.
2	Implement Player and Boss classes; test damage logic and gold system.
3	Finalize model logic; create beta GUI layout. Implement rough codes for all other classes
4	Test and refine attack flow; prepare a simple non-functional GUI.
5	Test the GUI (view) and display live HP updates.
6	Connect event handling (controller) for attack and upgrade buttons.
7	Debug and polish stage progression and death/reset behavior.
8	Finalize all documentation and code; thorough testing

GitHub Repository

<https://github.com/anhpls/epicbrawlers>